Exercise 4

Spherical collapse

All numbers, tables and figures below are produced by `notebooks/exercise_4.ipynb` with the code in `scripts/exercise_4.py`, following the smooth-dark-energy, shear-free limit of Pace, Meyer & Bartelmann [1] and the Lecture 8 notes [2]. The background is EdS or flat ΛCDM, $E^2(a) = \Omega_{m0} a^{-3} + \Omega_{\Lambda 0}$ with $\Omega_{\Lambda 0} = 1 - \Omega_{m0}$, with radiation neglected at all epochs.

1 Formulations A, B, C

For $N = \ln a$, $A_H = 2 + \mathrm{d}\ln H/\mathrm{d}N$, $\vartheta = \nabla_x \cdot \boldsymbol{v}/(aH)$, $g = 1/\delta$,

$$\text{A:}\quad \frac{d^2\delta}{dN^2} = -A_H \frac{d\delta}{dN} + \frac{4}{3}\frac{(\frac{d\delta}{dN})^2}{(1+\delta)} + \frac{3}{2}\Omega_m \delta(1+\delta), \tag{1}$$

$$\text{B:}\quad \frac{\mathrm{d}\delta}{\mathrm{d}N} = -(1+\delta)\,\vartheta, \qquad \frac{\mathrm{d}\vartheta}{\mathrm{d}N} = -A_H\,\vartheta - \frac{\vartheta^2}{3} - \frac{3}{2}\,\Omega_m\,\delta, \tag{2}$$

$$\text{C:}\quad \frac{\mathrm{d}g}{\mathrm{d}N} = g\,(1+g)\,\vartheta, \qquad \frac{\mathrm{d}\vartheta}{\mathrm{d}N} = -A_H\,\vartheta - \frac{\vartheta^2}{3} - \frac{3\,\Omega_m}{2g}. \tag{3}$$

A is integrated as the first-order system (δ, δ_N), B and C directly.

The same state (N, δ, ϑ) is written in each formulation's variables ($\delta_N = -(1+\delta)\vartheta$ for A, $g = 1/\delta$ for C) and compared through B:

- A vs B: δ^A_{NN} vs $-\delta_N\vartheta - (1+\delta)\vartheta^B_N$
- C vs B: $-\delta^2 g^C_N$ vs δ^B_N, and ϑ^C_N vs ϑ^B_N

Over 20 random states ($N \in [-10, 0]$, $\delta \in [10^{-5}, 10^{6}]$), the maximum $|x_1/x_2 - 1|$ is 3.3×10^{-16}, 4.4×10^{-16}, 2.2×10^{-16} (EdS) and 3.3×10^{-16}, 2.2×10^{-16}, 2.2×10^{-16} ($\Omega_{m0} = 0.3$) for the three comparisons.

2 Growing-mode initial conditions

$\delta_{N,i} = f_i \delta_i$ (A), $\vartheta_i = -f_i\delta_i/(1+\delta_i)$ (B, C), $g_i = 1/\delta_i$ (C), and for the linear equation $\delta_{L,NN} + A_H \delta_{L,N} - \frac{3}{2}\Omega_m \delta_L = 0$, integrated in parallel with each formulation, $\delta_{L,i} = \delta_i$, $\delta_{L,N,i} = f_i \delta_i$.

f_i is calculated from Exercise 1 at a_i (with ODE evaluated starting at $10^{-3} a_i$).

Table 1: $1 - f_i$ at the initial epoch (expected: $f_i = 1$ in EdS, and $f_i \to 1$ in ΛCDM for early a_i, matter domination).

Ω_{m0}	$a_i = 10^{-3}$	$a_i = 10^{-4}$	$a_i = 10^{-5}$
1.0	-2.2×10^{-16}	2.2×10^{-16}	-8.9×10^{-16}
0.2	2.18×10^{-9}	2.18×10^{-12}	1.3×10^{-15}
0.3	1.27×10^{-9}	1.27×10^{-12}	1.6×10^{-15}
0.4	8.18×10^{-10}	8.19×10^{-13}	5.6×10^{-16}

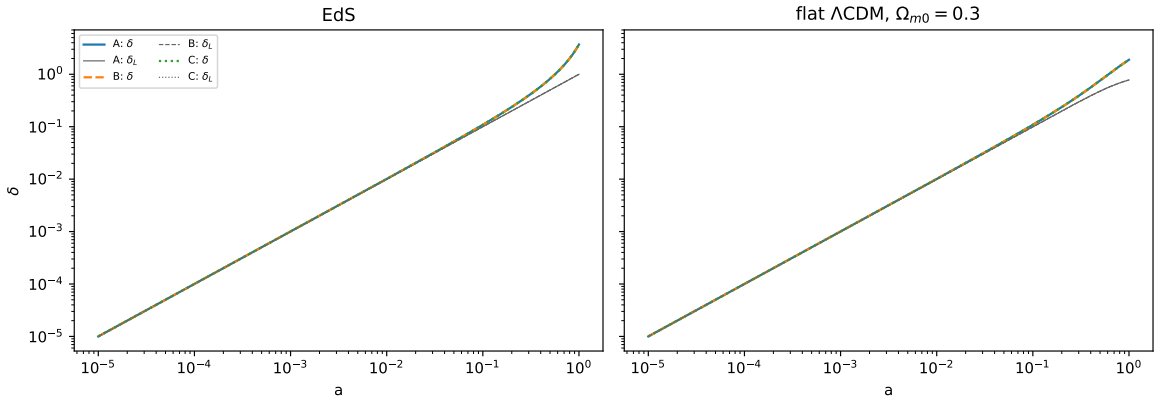


Figure 1: A, B, C integrated together with their linear counterpart for one trial amplitude, $\delta_i = 10^{-5}$ at $a_i = 10^{-5}$, from a_i to $a = 1$.

3 Collapse epoch

For $a_c = (1 + z_c)^{-1}$, δ_i is tuned with `brentq` until the collapse event ($\delta = \delta_\infty$ for A, B; $g = \epsilon$ for C) happens at a_c; then $\delta_c = \delta_L(a_c)$.

Residual: $N_{\rm coll}(\delta_i) - N_c$ (integration to $N_c + 0.5$). Bracket: $\pm 3\%$ around $1.686\, D(a_i)/D(a_c)$, widened until the sign changes.

For $a_i = 10^{-5}$, $\delta_\infty = 10^8$ (A, B), $\epsilon = 10^{-8}$ (C), ODEs are solved with method DOP853 with `rtol=1e-10`, `atol=1e-13`. The collapse event is terminal, so the singular endpoint is never crossed; `brentq` uses `xtol`$= 10^{-14}\times$guess, `rtol`$= 10^{-14}$.

Table 2: Tuned runs: root-search record and settings ($z_c = 0$; $a_i = 10^{-5}$, `rtol=1e-10`, `atol=1e-13`). Event residual $= \ln(a_{\rm coll}/a_c)$; nfev: right-hand-side evaluations of the tuned run.

	Ω_{m0}	threshold	$10^5\,\delta_i$	δ_c	bracket $[10^{-5}]$	iter.	event residual	nfev
A	1.0	$\delta_\infty = 10^8$	1.6863898	1.6863898	[1.63588, 1.73706]	6	4.5×10^{-15}	2357
B	1.0	$\delta_\infty = 10^8$	1.6863898	1.6863898	[1.63588, 1.73706]	6	-4.2×10^{-15}	2483
C	1.0	$\epsilon = 10^{-8}$	1.6863898	1.6863898	[1.63588, 1.73706]	6	4.5×10^{-15}	3101
A	0.3	$\delta_\infty = 10^8$	2.1508360	1.6754604	[2.10002, 2.22992]	7	-8.3×10^{-15}	2297
B	0.3	$\delta_\infty = 10^8$	2.1508360	1.6754604	[2.10002, 2.22992]	7	1.9×10^{-15}	2339
C	0.3	$\epsilon = 10^{-8}$	2.1508360	1.6754604	[2.10002, 2.22992]	7	-8.0×10^{-16}	2942

The 3 formulations A, B, C reach the threshold at a_c with $|\ln(a_{\rm coll}/a_c)| \lesssim 10^{-14}$ and return the same δ_c. As the threshold grows, δ_c increases.

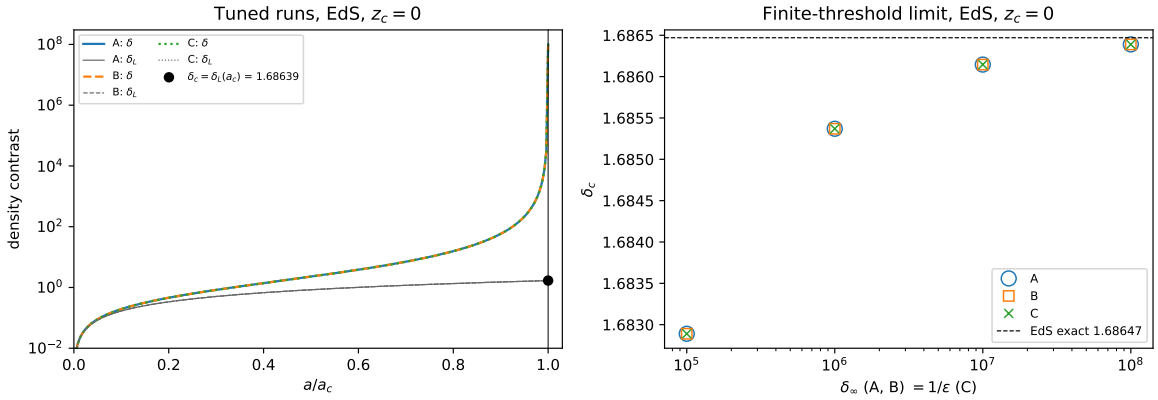


Figure 2: Left: tuned trajectories (EdS, $z_c = 0$) up to the collapse event and δ_c read at a_c. Right: δ_c for each threshold, $\delta_\infty = 10^{5\ldots8}$ (A, B) and $\epsilon = 1/\delta_\infty$ (C), δ_i retuned for each.

4 Turnaround and virial prescription

Turnaround: $\vartheta(a_{\rm ta}) = -3, \quad \zeta_{\rm ta} = 1 + \delta(a_{\rm ta})$, with $\vartheta = -\delta_N/(1+\delta)$ in A and $\delta = 1/g$ in C. It is recorded as a non-terminal event on the tuned run (from $\mathrm{d}\ln R/\mathrm{d}N = 1 + \vartheta/3$).

Virial prescription:

$$\lambda_{\rm ta} = \frac{\Omega_\Lambda(a_{\rm ta})}{\Omega_m(a_{\rm ta})\,\zeta_{\rm ta}}, \qquad 4\lambda_{\rm ta} y^3 - 2(1+\lambda_{\rm ta})\,y + 1 = 0, \quad y = \frac{R_{\rm vir}}{R_{\rm ta}} \in (0, 1/2] \quad (y = 1/2 \text{ in EdS}) \tag{4}$$

$$\Delta_{{\rm vir},m}(a_c) = \zeta_{\rm ta}\left(\frac{a_c}{a_{\rm ta}}\right)^3 y^{-3}, \qquad \Delta_{{\rm vir},c}(a_c) = \Omega_m(a_c)\,\Delta_{{\rm vir},m}(a_c) \tag{5}$$

The physical root of eq. (4) is found with `brentq` in $(0, 1/2]$: $y = 0.4999999975$ for $\lambda_{\rm ta} = 10^{-8}$, 0.4974876 for 10^{-2}, 0.4845798 for 0.06, 0.4738957 for 0.1, with cubic residuals $\leq 6 \times 10^{-16}$.

Table 3: Required outputs of each formulation ($z_c = 0$, default settings). A, B and C agree to all digits shown; residuals and costs of each formulation are in Table 2.

	EdS	$\Omega_{m0} = 0.3$
tuned δ_i	1.6863898×10^{-5}	2.1508360×10^{-5}
$a_{\rm ta}$ $(= a_{\rm ta}/a_c)$	0.6300026	0.5614617
$\zeta_{\rm ta}$	5.5516525	6.8373077
δ_c	1.6863898	1.6754604
$\lambda_{\rm ta}$	0	0.0604021
y	0.5	0.4844738
$\Delta_{{\rm vir},m}$	177.61729	339.71423
$\Delta_{{\rm vir},c}$	177.61729	101.91427

A, B, C return same turnaround and virial values.
For EdS: $a_{\rm ta}/a_c = 0.6300$, $\zeta_{\rm ta} = 5.5517$, $y = 1/2$, $\Delta_{{\rm vir},m} = \Delta_{{\rm vir},c} = 177.62$
For $\Omega_{m0} = 0.3$: $a_{\rm ta}/a_c = 0.5615$, $\zeta_{\rm ta} = 6.837$, $y = 0.4845 < 1/2$, $\Delta_{{\rm vir},m} = 339.7$, $\Delta_{{\rm vir},c} = 101.9$

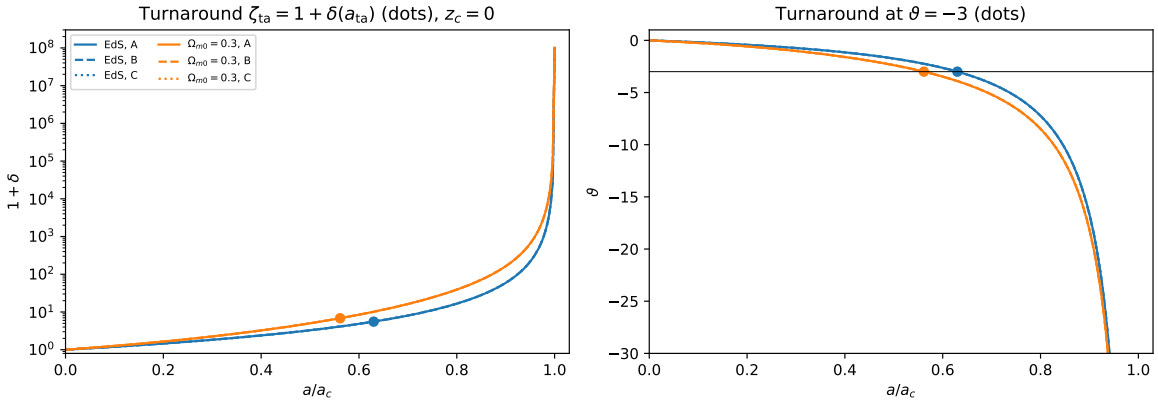


Figure 3: Tuned runs ($z_c = 0$) of A, B, C for EdS and $\Omega_{m0} = 0.3$. Left: $1 + \delta$ with the turnaround $(a_{\rm ta}, \zeta_{\rm ta})$ (dots). Right: ϑ, crossing -3 at turnaround (dots).

5 Independent EdS benchmarks

For $\delta_c = \frac{3}{20}(12\pi)^{2/3} = 1.68647$, $a_{\rm ta}/a_c = 2^{-2/3}$, $\zeta_{\rm ta} = (3\pi/4)^2$, $\Delta_{{\rm vir},m} = 18\pi^2$ at $z_c = 0, 1, 2$.

Table 4: Relative error of each formulation with respect to the exact EdS values (default settings).

z_c		δ_i	δ_c	$a_{\rm ta}/a_c$	$\zeta_{\rm ta}$	$\Delta_{{\rm vir},m}$
0	A	1.69×10^{-5}	-4.77×10^{-5}	6.68×10^{-5}	3.2×10^{-11}	-2.00×10^{-4}
	B	1.69×10^{-5}	-4.77×10^{-5}	6.68×10^{-5}	5.6×10^{-12}	-2.00×10^{-4}
	C	1.69×10^{-5}	-4.77×10^{-5}	6.68×10^{-5}	-7.9×10^{-11}	-2.00×10^{-4}
1	A	3.37×10^{-5}	-2.86×10^{-5}	6.68×10^{-5}	5.1×10^{-11}	-2.00×10^{-4}
	B	3.37×10^{-5}	-2.86×10^{-5}	6.68×10^{-5}	8.1×10^{-12}	-2.00×10^{-4}
	C	3.37×10^{-5}	-2.86×10^{-5}	6.68×10^{-5}	-1.4×10^{-10}	-2.00×10^{-4}
2	A	5.06×10^{-5}	-9.45×10^{-6}	6.68×10^{-5}	-2.3×10^{-12}	-2.00×10^{-4}
	B	5.06×10^{-5}	-9.45×10^{-6}	6.68×10^{-5}	3.8×10^{-12}	-2.00×10^{-4}
	C	5.06×10^{-5}	-9.45×10^{-6}	6.68×10^{-5}	1.1×10^{-10}	-2.00×10^{-4}

Convergence with the collapse threshold (B, $z_c = 0$) is shown in Table 5.

Table 5: Relative error with respect to the exact EdS values as a function of the collapse threshold (B, $z_c = 0$).

δ_∞	δ_c	$a_{\rm ta}/a_c$	$\zeta_{\rm ta}$	$\Delta_{{\rm vir},m}$
10^5	-2.12×10^{-3}	2.15×10^{-3}	5.0×10^{-12}	-6.41×10^{-3}
10^6	-6.53×10^{-4}	6.72×10^{-4}	5.5×10^{-12}	-2.01×10^{-3}
10^7	-1.92×10^{-4}	2.12×10^{-4}	5.6×10^{-12}	-6.35×10^{-4}
10^8	-4.77×10^{-5}	6.68×10^{-5}	5.6×10^{-12}	-2.00×10^{-4}

The turnaround checks ($a_{\rm ta}/a_c$, $\zeta_{\rm ta}$) and $\Delta_{{\rm vir},m}$ improve together with δ_c as the threshold grows, so the benchmarks are not recovered by tuning the threshold to δ_c alone.

6 Cosmology grid

$\Omega_{m0} \in \{0.2, 0.3, 0.4\}$, $\Omega_{\Lambda 0} = 1 - \Omega_{m0}$, $z_c \in \{0, 1, 2\}$, default settings, formulations A, B, C. The full table of the 27 runs is saved in `data/lcdm_grid.csv`.

Growth: δ_c is the initial amplitude needed for collapse at a_c, evolved in linear theory from a_i to a_c. In ΛCDM, cosmologies with higher values of Λ have slower growth of linear perturbations at late times, so for an overdensity to collapse at a_c, it needs larger initial amplitudes than in EdS.

Table 6: Grid results (formulation C). The cross-formulation spread (max − min over A, B, C) is $\leq 1.0 \times 10^{-10}$ in δ_c, $\leq 7.9 \times 10^{-11}$ in $a_{\rm ta}/a_c$, $\leq 2.0 \times 10^{-9}$ in $\zeta_{\rm ta}$, $\leq 1.3 \times 10^{-7}$ in $\Delta_{{\rm vir},m}$ and $\leq 2.5 \times 10^{-8}$ in $\Delta_{{\rm vir},c}$ at every grid point.

	δ_c			$\Delta_{{\rm vir},m}$			$\Delta_{{\rm vir},c}$		
Ω_{m0}	$z_c = 0$	1	2	0	1	2	0	1	2
0.2	1.67161	1.68289	1.68527	440.00	216.51	189.57	88.00	144.34	165.11
0.3	1.67546	1.68421	1.68575	339.71	200.78	184.64	101.91	155.44	169.96
0.4	1.67817	1.68495	1.68599	286.19	192.68	182.15	114.47	162.25	172.56

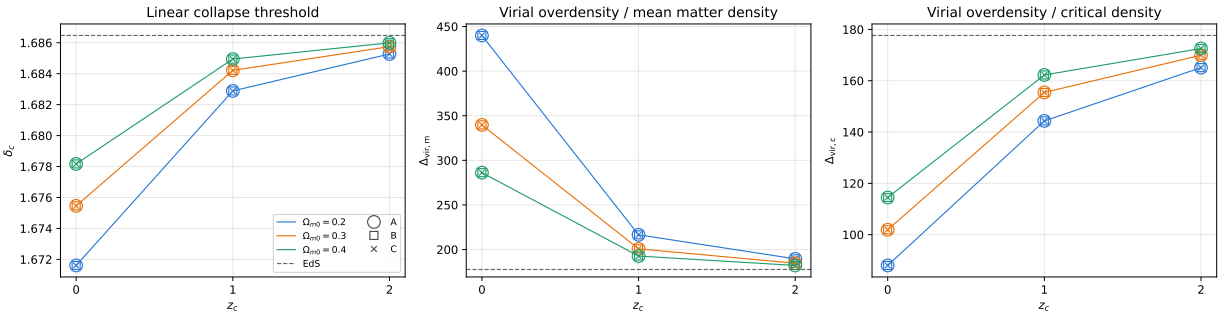


Figure 4: $\delta_c(z_c)$, $\Delta_{{\rm vir},m}(z_c)$ and $\Delta_{{\rm vir},c}(z_c)$ for the flat-ΛCDM grid; markers: A, B, C; dashed: EdS baselines 1.68647 and $18\pi^2$.

Table 7: $\delta_c = \delta_i\, D(a_c)/D(a_i)$: initial amplitude and linear growth in ΛCDM relative to EdS at the same z_c (formulation C).

Ω_{m0}	z_c	growth ΛCDM / growth EdS	δ_i ΛCDM / EdS	δ_c ΛCDM / EdS
0.2	0	0.70687	1.40229	0.99123
0.2	1	0.92561	1.07810	0.99790
0.2	2	0.97479	1.02514	0.99930
0.3	0	0.77898	1.27541	0.99352
0.3	1	0.95317	1.04776	0.99869
0.3	2	0.98489	1.01491	0.99958
0.4	0	0.83140	1.19692	0.99513
0.4	1	0.96861	1.03150	0.99913
0.4	2	0.99015	1.00967	0.99973

Table 8: Factorisation of the virial overdensities relative to EdS: $\Delta_{{\rm vir},m}/18\pi^2 = [\zeta_{\rm ta}/(3\pi/4)^2] \times [(a_c/a_{\rm ta})^3/4] \times [(1/2)/y]^3$, $\Delta_{{\rm vir},c} = \Omega_m(a_c)\Delta_{{\rm vir},m}$ (formulation C).

Ω_{m0}	z_c	$\Omega_m(a_c)$	$\dfrac{\zeta_{\rm ta}}{\zeta_{\rm ta}({\rm EdS})}$	$\dfrac{(a_c/a_{\rm ta})^3}{(a_c/a_{\rm ta})^3({\rm EdS})}$	$\dfrac{y^{-3}}{y^{-3}({\rm EdS})}$	$\dfrac{\Delta_{{\rm vir},m}}{\Delta_{{\rm vir},m}({\rm EdS})}$	$\dfrac{\Delta_{{\rm vir},c}}{\Delta_{{\rm vir},c}({\rm EdS})}$
0.2	0	0.200	1.342	1.618	1.141	2.477	0.495
0.2	1	0.667	1.065	1.112	1.029	1.219	0.812
0.2	2	0.871	1.021	1.036	1.010	1.067	0.929
0.3	0	0.300	1.232	1.412	1.099	1.912	0.574
0.3	1	0.774	1.039	1.068	1.018	1.130	0.875
0.3	2	0.920	1.012	1.021	1.006	1.039	0.957
0.4	0	0.400	1.165	1.290	1.072	1.611	0.644
0.4	1	0.842	1.026	1.045	1.012	1.085	0.913
0.4	2	0.947	1.008	1.013	1.004	1.025	0.971

Turnaround: the spheres of overdensities reach turnaround when they stop expanding due to their self-gravity. In ΛCDM, higher values of Λ make the background expansion accelerate, which means that overdensities need to reach higher values for turnaround. Indeed, $\zeta_{\rm ta}$ is larger for smaller values of Ω_m. For higher values of Λ, it also happens earlier in relation to the collapse, which makes $\Delta_{{\rm vir},m}$ increase.

Virial prescription: For a uniform sphere of mass M and radius R in a flat background with matter and a cosmological constant Λ, the virial prescription is defined based on the assumptions:

- For the gravitational potential energy of the sphere $U_G = -\dfrac{3GM^2}{5R}$ and the potential energy of the sphere in the Λ field $U_\Lambda = -\dfrac{\Lambda M R^2}{10}$, turnaround happens when the sphere is at rest ($K = 0$) and $E_{\rm ta} = U_{G,{\rm ta}} + U_{\Lambda,{\rm ta}}$
- It virializes when $2K + U_G - 2U_\Lambda = 0 \;\Rightarrow\; E_{\rm vir} = \frac{1}{2}U_{G,{\rm vir}} + 2U_{\Lambda,{\rm vir}}$
- Imposing energy conservation $E_{\rm ta} = E_{\rm vir}$, with $y = R_{\rm vir}/R_{\rm ta}$ and $\lambda_{\rm ta} = \dfrac{\rho_\Lambda}{\rho_{\rm sph,ta}} = \dfrac{\Omega_\Lambda(a_{\rm ta})}{\Omega_m(a_{\rm ta})\,\zeta_{\rm ta}}$:

$$4\lambda_{\rm ta}\, y^3 - 2(1+\lambda_{\rm ta})\, y + 1 = 0, \qquad y \in (0, 1/2] \quad (y = 1/2 \text{ in EdS})$$

- The density of the virialized halo follows from the density at turnaround as:

$$\rho_{\rm vir} = \rho_{\rm sph,ta}\left(\frac{R_{\rm ta}}{R_{\rm vir}}\right)^3 = \zeta_{\rm ta}\,\bar\rho_m(a_{\rm ta})\, y^{-3}$$

The virial overdensity is defined by comparing $\rho_{\rm vir}$ with the background at the a_c when the ideal sphere would reach $R = 0$. Relative to the mean matter density, which has diluted as a^{-3} since turnaround, $\bar\rho_m(a_c) = \bar\rho_m(a_{\rm ta})\,(a_{\rm ta}/a_c)^3$, and relative to the critical density, $\rho_{\rm crit} = \bar\rho_m/\Omega_m$:

$$\Delta_{{\rm vir},m}(a_c) = \frac{\rho_{\rm vir}}{\bar\rho_m(a_c)} = \zeta_{\rm ta}\left(\frac{a_c}{a_{\rm ta}}\right)^3 y^{-3}, \qquad \Delta_{{\rm vir},c}(a_c) = \frac{\rho_{\rm vir}}{\rho_{\rm crit}(a_c)} = \Omega_m(a_c)\,\Delta_{{\rm vir},m}(a_c)$$

For larger Λ, then, halos reach equilibrium at smaller radii and thus higher densities, which increases both $\Delta_{\rm vir,m}$ and $\Delta_{\rm vir,c}$. The virial prescription affects $\Delta_{\rm vir,m}$, $\Delta_{\rm vir,c}$ and y, values defined to describe dynamics and an equilibrium condition after the turnaround that the equations for δ_c do not, since they describe a spherical collapse to a point.

Reference density: $\Delta_{\rm vir,m}$ and $\Delta_{\rm vir,c}$ describe the halo density $\rho_{\rm vir}$ compared with the mean matter density $\bar\rho_m$ and the critical density $\rho_{\rm crit} = \bar\rho_m + \rho_\Lambda$. Because $\bar\rho_m$ dilutes as a^{-3}, while ρ_Λ stays constant, at late times $\rho_{\rm crit}$ is much larger than $\bar\rho_m$ ($\Omega_m(a_c) = \bar\rho_m/\rho_{\rm crit} < 1$). For a same halo, then, towards low z_c, $\Delta_{\rm vir,m}$ grows while $\Delta_{\rm vir,c}$ decreases. In EdS, where $\Lambda = 0$, the two reference densities are equal, as for at high z_c, when Λ is still negligible.

7 Convergence

Each setting is changed one at a time from the defaults ($a_i = 10^{-5}$, `rtol=1e-10`, `atol=1e-13`, $\delta_\infty = 10^8$, $\epsilon = 10^{-8}$), and δ_i is retuned for every run. Cases: EdS and flat ΛCDM $\Omega_{m0} = 0.3$, both at $z_c = 0$. "spread" = max − min over A, B, C; nfev = right-hand-side evaluations of the tuned run.

Table 9: Initial epoch a_i (A, B, C agree on δ_c to the digits shown).

Ω_{m0}	a_i	δ_c	spread δ_c	$\Delta_{\mathrm{vir},m}$	spread $\Delta_{\mathrm{vir},m}$	max event residual	nfev A / B / C
1.0	10^{-3}	1.6895862	1.0×10^{-10}	177.61729	3.1×10^{-8}	3.0×10^{-15}	2207 / 2333 / 2636
1.0	10^{-4}	1.6866799	9.7×10^{-11}	177.61729	2.4×10^{-8}	3.3×10^{-15}	2297 / 2423 / 2876
1.0	10^{-5}	1.6863898	9.8×10^{-11}	177.61729	7.3×10^{-9}	4.5×10^{-15}	2357 / 2483 / 3101
0.3	10^{-3}	1.6795126	1.0×10^{-10}	339.71423	8.1×10^{-8}	1.2×10^{-14}	2132 / 2189 / 2465
0.3	10^{-4}	1.6758280	1.0×10^{-10}	339.71423	1.4×10^{-8}	4.4×10^{-15}	2222 / 2279 / 2690
0.3	10^{-5}	1.6754604	1.0×10^{-10}	339.71423	6.6×10^{-8}	8.3×10^{-15}	2297 / 2339 / 2942

Table 10: Solver tolerances.

Ω_{m0}	rtol	atol	δ_c^A	δ_c^B	δ_c^C	spread δ_c	spread $\Delta_{\mathrm{vir},m}$	max event res.	nfev A / B / C
1.0	10^{-6}	10^{-9}	1.6863928	1.6863923	1.6863904	2.3×10^{-6}	2.8×10^{-4}	3.3×10^{-15}	1184 / 1022 / 1019
1.0	10^{-8}	10^{-11}	1.6863898	1.6863898	1.6863898	1.2×10^{-8}	1.7×10^{-6}	1.2×10^{-14}	2273 / 1802 / 1760
1.0	10^{-10}	10^{-13}	1.6863898	1.6863898	1.6863898	9.8×10^{-11}	7.3×10^{-9}	4.5×10^{-15}	2357 / 2483 / 3101
1.0	10^{-12}	10^{-15}	1.6863898	1.6863898	1.6863898	3.5×10^{-13}	2.1×10^{-10}	3.3×10^{-15}	4472 / 3467 / 3722
1.0	10^{-10}	10^{-9}	1.6863899	1.6863899	1.6863898	1.0×10^{-7}	7.4×10^{-8}	6.9×10^{-15}	2387 / 2489 / 3008
1.0	10^{-10}	10^{-6}	1.6864426	1.6864324	1.6863899	5.3×10^{-5}	2.9×10^{-5}	9.9×10^{-14}	2090 / 1979 / 2060
0.3	10^{-6}	10^{-9}	1.6754630	1.6754629	1.6754613	1.6×10^{-6}	9.4×10^{-4}	2.4×10^{-14}	1142 / 968 / 953
0.3	10^{-8}	10^{-11}	1.6754604	1.6754604	1.6754604	1.0×10^{-8}	3.9×10^{-6}	1.3×10^{-14}	2207 / 1748 / 1667
0.3	10^{-10}	10^{-13}	1.6754604	1.6754604	1.6754604	1.0×10^{-10}	6.6×10^{-8}	8.3×10^{-15}	2297 / 2339 / 2942
0.3	10^{-12}	10^{-15}	1.6754604	1.6754604	1.6754604	3.8×10^{-13}	2.4×10^{-10}	4.4×10^{-15}	4352 / 3377 / 3602
0.3	10^{-10}	10^{-9}	1.6754605	1.6754605	1.6754604	9.5×10^{-8}	1.6×10^{-7}	6.8×10^{-15}	2279 / 2396 / 2861
0.3	10^{-10}	10^{-6}	1.6755110	1.6755016	1.6754605	5.0×10^{-5}	1.3×10^{-4}	1.2×10^{-12}	2018 / 1925 / 1910

Table 11: Collapse threshold ($\epsilon = 1/\delta_\infty$ for C).

Ω_{m0}	δ_∞	ϵ	δ_c^B	spread δ_c	a_{ta}/a_c (B)	$\Delta_{\mathrm{vir},m}$ (B)	max event res.	nfev A / B / C
1.0	10^5	10^{-5}	1.6828900	9.7×10^{-11}	0.6313128	176.51376	5.7×10^{-15}	1697 / 1769 / 2399
1.0	10^6	10^{-6}	1.6853694	9.8×10^{-11}	0.6303840	177.29508	5.5×10^{-15}	1907 / 2006 / 2642
1.0	10^7	10^{-7}	1.6861456	9.7×10^{-11}	0.6300939	177.54013	1.8×10^{-15}	2132 / 2258 / 2858
1.0	10^8	10^{-8}	1.6863898	9.8×10^{-11}	0.6300026	177.61729	4.5×10^{-15}	2357 / 2483 / 3101
0.3	10^5	10^{-5}	1.6721165	1.0×10^{-10}	0.5631211	337.50725	5.9×10^{-15}	1622 / 1613 / 2213
0.3	10^6	10^{-6}	1.6744872	1.0×10^{-10}	0.5619433	339.07104	5.3×10^{-15}	1847 / 1877 / 2456
0.3	10^7	10^{-7}	1.6752277	1.0×10^{-10}	0.5615768	339.56036	4.8×10^{-15}	2072 / 2102 / 2699
0.3	10^8	10^{-8}	1.6754604	1.0×10^{-10}	0.5614617	339.71423	8.3×10^{-15}	2297 / 2339 / 2942

Table 12: Failure modes: loose tolerances, EdS, $z_c = 0$. Relative shift of δ_c with respect to the default run of each formulation, and cost.

		δ_c shift			nfev		
rtol	atol	A	B	C	A	B	C
10^{-4}	10^{-13}	1.03×10^{-4}	7.26×10^{-5}	1.12×10^{-4}	737	656	650
10^{-10}	10^{-5}	3.21×10^{-4}	2.12×10^{-4}	6.8×10^{-7}	1955	1760	1601
10^{-10}	10^{-4}	3.67×10^{-4}	2.72×10^{-4}	2.7×10^{-6}	1835	1589	1226

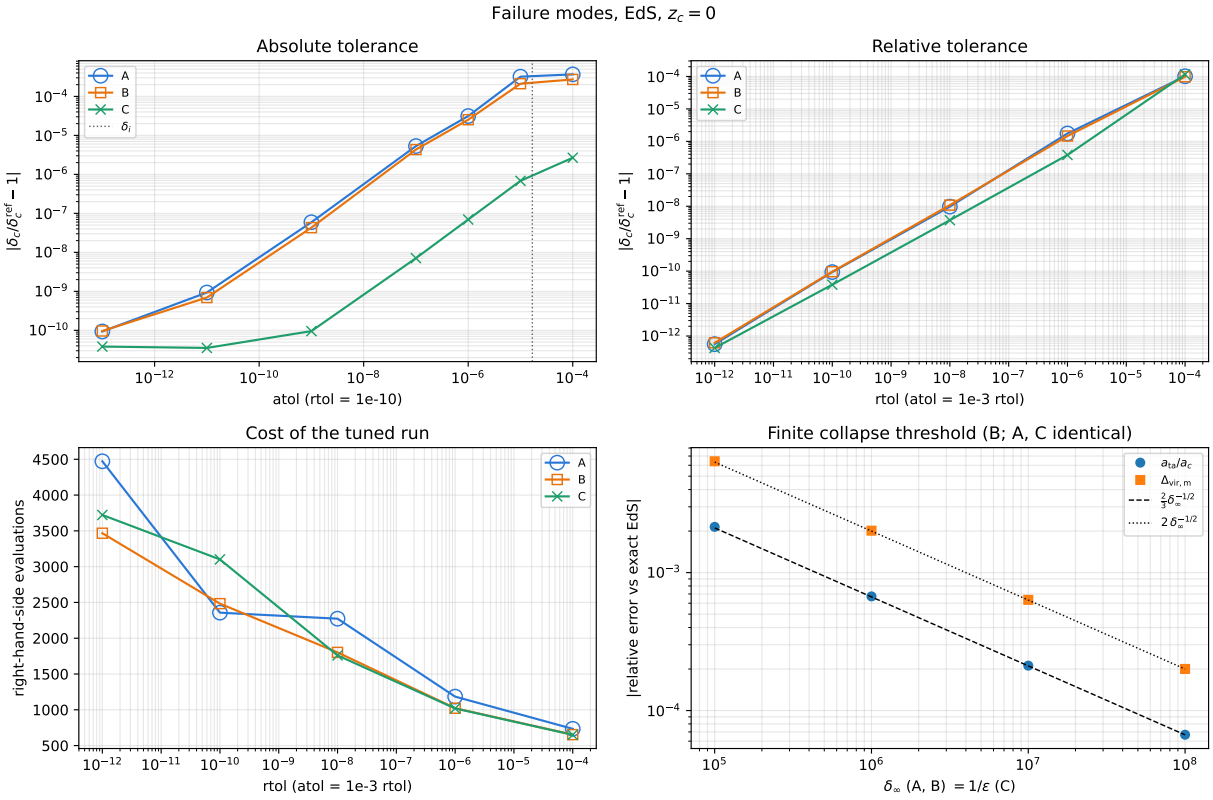


Figure 5: Failure modes, EdS, $z_c = 0$. Top: error of δ_c against the tightest run (`rtol=1e-13`, `atol=1e-16`) of each formulation, as a function of `atol` (`rtol=1e-10`) and `rtol` (`atol`$= 10^{-3}$ `rtol`). Bottom left: cost of the tuned run. Bottom right: errors of $a_{\rm ta}/a_c$ and $\Delta_{{\rm vir},m}$ against the exact EdS values as a function of the threshold (B; A, C identical).

The figure separates the failure modes: A and B lose accuracy when `atol` approaches the initial amplitude δ_i, while C does not ($g = 1/\delta_i$ is large). `rtol` limits all three equally (error $\approx$ `rtol`), and A is the most expensive at the tightest tolerance. The threshold produces the same error in all formulations, $\propto \delta_\infty^{-1/2}$ and is the dominant error.

8 Code and reproduction

`scripts/exercise_4.py`: background and shared integration, shooting (`tune_delta_i`) and turnaround/virial step (`turnaround_virial`) in the abstract class `Background`; formulations `A`, `B`, `C` with their own equations, initial conditions and events; `virial_y` for the root of eq. (4). The growth rate f_i and the growth factor come from `Cosmology` in `scripts/exercise_1.py`. The notebook `notebooks/exercise_4.ipynb` produces all tables and figures and saves them in `report_ex4/data` and `report_ex4/figures`; see `report_ex4/README.md` for dependencies and commands.

References

[1] F. Pace, S. Meyer, M. Bartelmann, "On the implementation of the spherical collapse model for dark energy models", JCAP **10**, 040 (2017), arXiv:1708.02477.

[2] T. Castro, *Spherical Collapse: from the EdS solution to halo abundances*, Numerical Cosmology Lecture 8 notes.